This leads us to the **flat-top window**, shown in Fig. 9-5d. In some applications the *amplitude* of a spectral peak must be measured very accurately. Since the DFT's frequency spectrum is formed from samples, there is nothing to guarantee that a sample will occur exactly at the top of a peak. More than likely, the nearest sample will be slightly off-center, giving a value lower than the true amplitude. The solution is to use a window that produces a spectral peak with a *flat top*, insuring that one or more of the samples will always have the correct peak value. As shown in Fig. 9-5d, the penalty for this is a very broad main lobe, resulting in poor frequency resolution.

As it turns out, the shape we want for a flat-top window is exactly the same shape as the filter kernel of a low-pass filter. We will discuss the theoretical reasons for this in later chapters; for now, here is a cookbook description of how the technique is used. Chapter 16 discusses a low-pass filter called the *windowed-sinc*. Equation 16-4 describes how to generate the filter kernel (which we want to use as a window), and Fig. 16-4a illustrates the typical shape of the curve. To use this equation, you will need to know the value of two parameters: M and f_c. These are found from the relations: $M = N - 2$, and $f_c = s/N$, where N is the length of the DFT being used, and s is the number of samples you want on the flat portion of the peak (usually between 3 and 5). Table 16-1 shows a program for calculating the filter kernel (our window), including two subtle features: the normalization constant, K, and how to avoid a divide-by-zero error on the center sample. When using this method, remember that a DC value of *one* in the time domain will produce a peak of amplitude *one* in the frequency domain. However, a sinusoid of amplitude *one* in the time domain will only produce a spectral peak of amplitude *one-half*. (This is discussed in the last chapter: *Synthesis, Calculating the Inverse DFT*).

Frequency Response of Systems

Systems are analyzed in the *time domain* by using convolution. A similar analysis can be done in the *frequency domain.* Using the Fourier transform, every input signal can be represented as a group of cosine waves, each with a specified amplitude and phase shift. Likewise, the DFT can be used to represent every output signal in a similar form. This means that any linear system can be *completely* described by how it changes the amplitude and phase of cosine waves passing through it. This information is called the system's **frequency response.** Since both the impulse response and the frequency response contain complete information about the system, there must be a one-to-one correspondence between the two. Given one, you can calculate the other. The relationship between the impulse response and the frequency response is one of the foundations of signal processing: *A system's frequency response is the Fourier Transform of its impulse response*. Figure 9-6 illustrates these relationships.

Keeping with standard DSP notation, impulse responses use lower case variables, while the corresponding frequency responses are upper case. Since $h[\ ]$ is the common symbol for the impulse response, $H[\ ]$ is used for the frequency response. Systems are described in the time domain by convolution, that is: